

Ex.2

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```
print(uuid::UUIDgenerate())
```

```
## [1] "3d3b00a0-5dd6-4779-82b0-b23aa2dc64ab"
```

Q1

```
commute <- c(18, 25, 32, 41, 27, 35, 22, 19, 45, 38, 29, 24, 31, 36, 40, 26, 33, 28, 21, 44, 30, 37, 23)
max(commute);min(commute)
```

```
## [1] 45
```

```
## [1] 18
```

$$(45-18)/5$$

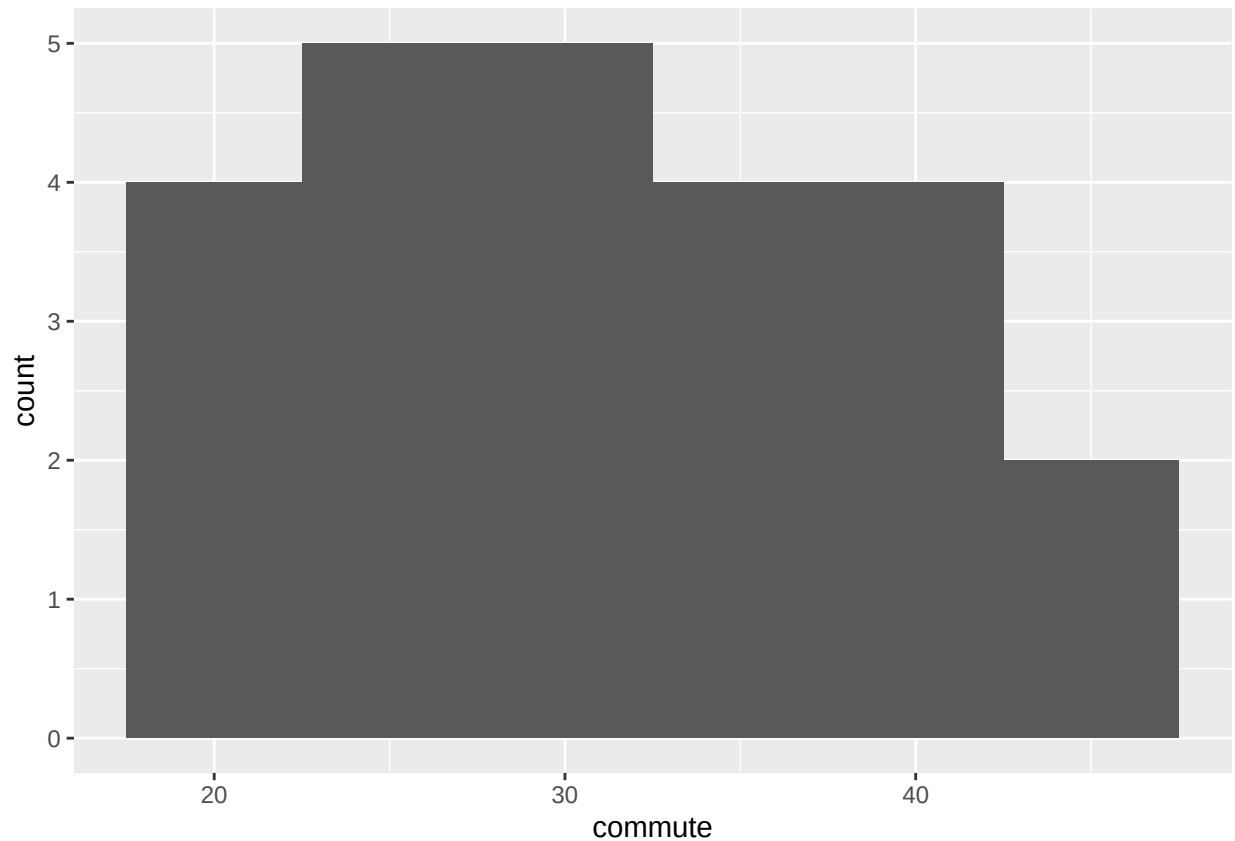
```
## [1] 5.4
```

```
table(cut(commute,5))
```

```
##
## (18,23.4] (23.4,28.8] (28.8,34.2] (34.2,39.6] (39.6,45]
##          5          5          5          5          4
```

Q2

```
library(ggplot2)
#Draw histogram
ggplot(data.frame(commute),aes(x=commute))+
  geom_histogram(binwidth=5)
```



Q3

```
## Q3

# Divide into 5 classes
a <- cut(commute, 5)

# Frequency
freq <- table(a)

# Cumulative frequency
cum_freq <- cumsum(freq)

# Upper class boundaries
upper_boundaries <- c(23.4, 28.8, 34.2, 39.6, 45)

# Create data frame
ogive_data <- data.frame(
  upper_boundary = upper_boundaries,
  cumulative_frequency = cum_freq
)

# Draw ogive
ggplot(ogive_data, aes(x = upper_boundary, y = cumulative_frequency)) +
```

```
geom_line() +  
geom_point()
```

